

Assignment A4

For feature 1 and feature 2 design for the yield strength, then design for the max deflection 0.30mm

Material: ABS

Force acting on motor: $P = 300\text{N}$

Safety factor: 3

ABS:

Elastic modulus: $290,075\text{ psi} = 2,000\text{ MPa}$

Yield strength = 40 MPa